

Perplexity is defined as the exponential of Cross entropy loss

$$\text{Cross-entropy loss: } - \sum_{i=1}^n \sum_{j=1}^k I(y_i=j) \log(\hat{y}_j)$$

$$\text{perplexity: } \exp(\text{Cross-entropy loss} / n)$$



perplexity is more interpretable than cross-entropy loss

1. Cross-entropy = 3.56 → hard to explain: what does this number mean

perplexity = 3.7 → meaning: in average, the model is confused with 3.7 equally likely words.

Why: Say there are k words, the model has no idea which one to pick and choose each word with equal probability $\frac{1}{k}$.

Cross-entropy for choosing N times

$$\begin{aligned} & - \sum_{i=1}^N \sum_{j=1}^k I(y_i=j) \log\left(\frac{1}{k}\right) \\ & = - \sum_{i=1}^N \log\left(\frac{1}{k}\right) \\ & = - \sum_{i=1}^N (\log(1) - \log(k)) \\ & = \sum_{i=1}^N \log(k) = N \log(k) \end{aligned}$$

$$\Rightarrow \underline{\text{perplexity} = \exp(N \log(k) / N) = k}$$

2. When

Cross-entropy goes down 10% $\rightarrow$ what does it mean?

perplexity from 69 $\rightarrow$ 51 $\rightarrow$ the model was confused with 69 words, and now confused with 51.